

n8n — AI Workflow Automation

No-code/low-code automation: connect any app to any AI model

01 What is n8n?

n8n (pronounced "n-eight-n") is an open-source workflow automation tool — like Zapier or Make, but self-hostable, with code execution, and first-class AI/LLM integration. You build workflows as visual flowcharts connecting nodes.

Feature	n8n	Zapier	Make (Integromat)
Self-hostable	Yes (Docker)	No (cloud only)	No (cloud only)
AI/LLM nodes	Built-in (LangChain)	Limited	Limited
Code execution	JavaScript / Python nodes	No	Limited
Vector store integration	Yes (Pinecone, Qdrant, Supabase)	No	No
Agent workflows	Yes (AI Agent node)	No	No
Cost	Free self-hosted	Paid per task	Paid per operation

02 Core n8n Concepts

Concept	What It Is	Example
Workflow	A connected sequence of nodes that automates a process	Email arrives → extract info → save to DB → notify Slack
Node	A single step in the workflow (trigger, action, or transformation)	HTTP Request, OpenAI, Gmail, Slack, Code, IF, Set
Trigger Node	Starts the workflow on an event	Webhook, Schedule (cron), Email received, Form submitted
Action Node	Performs an operation	Send email, query DB, call API, run code
AI Node	Calls LLMs or performs AI operations	OpenAI Chat, LangChain Agent, Embeddings, Vector Store
Data Flow	JSON data passed between nodes	Each node receives input items, outputs transformed items

03 Setting Up n8n

Run n8n with Docker (quickest)

```
docker run -it --rm \
  -p 5678:5678 \
  -v ~/.n8n:/home/node/.n8n \
  n8nio/n8n

# Open browser: http://localhost:5678
# Create account and start building workflows
```

```
# n8n with Docker Compose (production)
version: "3"
services:
  n8n:
    image: n8nio/n8n
    ports: ["5678:5678"]
    environment:
      - N8N_BASIC_AUTH_ACTIVE=true
      - N8N_BASIC_AUTH_USER=admin
      - N8N_BASIC_AUTH_PASSWORD=yourpassword
    volumes:
      - n8n_data:/home/node/.n8n
volumes:
  n8n_data:
```

04 AI Workflow Patterns

Pattern 1: Simple LLM Processing Pipeline

Trigger (Webhook/Form) → **Set node** (prepare data) → **OpenAI Chat** (LLM call) → **Send Email / Slack** (deliver result)

EXAMPLE USE CASE

Customer submits support ticket → n8n sends to GPT-4o with product context → LLM generates draft response → Human reviews → Auto-send if confidence > 0.9

Pattern 2: RAG Workflow

Trigger → **Embeddings node** (embed query) → **Vector Store node** (retrieve chunks) → **OpenAI Chat** (RAG prompt with chunks) → **Output**

n8n has built-in nodes for: Pinecone, Qdrant, Supabase pgvector, In-memory vector store. No code needed for basic RAG.

Pattern 3: AI Agent Workflow

The **AI Agent node** in n8n wraps LangChain's agent execution. Connect tools (Calculator, Wikipedia, HTTP Request, Code Exec) to the agent. The agent decides which tools to call based on the input.

Pattern 4: Document Ingestion Pipeline

Schedule trigger (daily) → **Google Drive** (list new files) → **Extract from File** (PDF text) → **Text Splitter** (chunk) → **Embeddings** → **Vector Store** (upsert) → **Slack notification**

05 Key n8n Nodes for AI Workflows

Node	Category	What It Does
OpenAI	AI	Chat completion, embeddings, image generation, Assistants API
AI Agent	AI	LangChain ReAct agent with configurable tools
Text Splitter	AI	Split documents: recursive, token, markdown header
Embeddings (OpenAI/HuggingFace)	AI	Generate embeddings for text
Vector Store (Pinecone/Qdrant)	AI	Insert and query vector databases
Retrieval Q&A; Chain	AI	RAG chain: embed query → retrieve → LLM answer
HTTP Request	Core	Call any REST API
Code	Core	Run JavaScript or Python in the workflow
IF	Core	Branch workflow based on conditions
Webhook	Trigger	Start workflow on HTTP POST from any source
Schedule Trigger	Trigger	Run workflow on cron schedule
Gmail / Outlook	Integration	Read/send emails
Slack / Teams	Integration	Send messages, read channels
PostgreSQL / MySQL	Integration	Query and update databases
Google Sheets	Integration	Read/write spreadsheet data

06 Practical AI Use Cases

Use Case	Workflow
AI Email Triage	Email received → classify urgency → route to right team → draft response
Auto-summarize Meetings	Zoom recording → Whisper transcription → GPT summary → save to Notion
Customer FAQ Bot	Chat trigger → embed question → Pinecone search → LLM answer with context
AI Code Review	GitHub PR webhook → extract diff → Claude review → post comment on PR
Content Generation Pipeline	RSS feed → extract articles → LLM summary → post to LinkedIn/Twitter
Invoice Processing	Email attachment → extract text → LLM parse fields → save to Airtable
Slack AI Assistant	Slack mention → RAG over company docs → reply in thread
KYC Document Check	Document upload → OCR → LLM extract fields → validate → flag anomalies